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Epistemic Cryptography 2.0

Canonical Structural Fingerprinting of Relational Fields

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Abstract

Epistemic Cryptography (EK) is a non-interventional framework for deriving deterministic structural fingerprints from relational systems without modifying their behavior. EK 2.0 extends the original formulation by introducing: canonicalization over symmetry orbits, representability constraints, topology-sensitive hashing, and trajectory verification via Merkle structures. The system operates purely on structural observables and introduces no optimization, no feedback, and no interpretative layer.

Keywords: Epistemic Cryptography, Structural Fingerprinting, Canonical Forms, Relational Geometry, $so(8)$, Merkle Verification, Non-Intervention

I. Ontological Domain

Let:

$$\Phi = (\Sigma, R)$$

Σ : finite index set, $|\Sigma| = 8$

$R \in \mathfrak{so}(8)$: antisymmetric relational matrix

Define the curvature vector:

$$\Delta = dR \in \mathbb{R}^{56}$$

Δ represents curvature derived from triangle inequality deviations over relational triples (i,j,k) . For $|\Sigma| = 8$, the number of independent triples is $C(8,3) = 56$, corresponding to the dimension of $\mathfrak{so}(8)$.

Explicitly:

$$\Delta(i,j,k) = R_{ij} + R_{jk} + R_{ki}$$

II. Representability Constraint

Define the admissible domain:

$$D = \{ \Delta \mid \exists R \in \mathfrak{so}(8), \Delta = dR \}$$

A valid epistemic field satisfies:

$$\Delta \in D$$

Representability is verified by reconstructing:

$$\hat{R} = \pi(\text{encodings})$$

$$\hat{\Delta} = d(\hat{R})$$

and requiring:

$$\hat{\Delta} \equiv \Delta$$

D is a non-convex, bounded, nonlinear submanifold of $\mathbb{R}^{56}$. Its boundary $dD = \kappa$ represents the limit of epistemic representability. Fields outside D are in Qualitative Epistemic Aporia (QE).

III. Symmetry Structure

The relational structure is invariant under relabeling of index set elements. The symmetry group is:

$$G = S_8 \text{ (symmetric group on 8 elements, } |S_8| = 40320)$$

Group action on Delta:

$$(g \cdot \text{Delta})(i, j, k) = \text{Delta}(g^{-1}(i), g^{-1}(j), g^{-1}(k))$$

Symmetry orbit:

$$O(\text{Delta}) = \{ g \cdot \text{Delta} \mid g \in S_8 \}$$

EK must be invariant over $O(\text{Delta})$. Two fields that differ only by relabeling are structurally identical and must produce identical fingerprints.

Note: Full orbit enumeration over S_8 is computationally intensive ($|S_8| = 40320$). Practical implementations use canonical form selection rather than explicit orbit traversal.

IV. Canonical Selection Rule

Equivalence relation:

$$\text{Delta}_1 \sim \text{Delta}_2 \text{ iff } \text{Delta}_2 \in O(\text{Delta}_1)$$

Define a deterministic canonical mapping:

$$C : O(\text{Delta}) \rightarrow \text{Delta}_c$$

such that:

$$C(g \cdot \text{Delta}) = C(\text{Delta}) \text{ for all } g \in S_8$$

Design constraints on C:

Deterministic: identical input always produces identical output

Invariant under G: $C(g \cdot \text{Delta}) = C(\text{Delta})$

No optimization objective: C does not minimize or maximize any function

No semantic ordering: selection criterion is purely structural

Canonicalization selects a unique representative per orbit. The selection criterion is a structural invariant — for example, lexicographic ordering of the sorted eigenvalue sequence of the reconstructed R matrix. This criterion is structural, not semantic.

V. Structural Pipeline

The full EK 2.0 pipeline:

```
system -> outputs -> encode -> Delta -> canonicalize -> kappa ->
H -> artifact
```

Pipeline steps:

1. Observe outputs (no intervention, no adaptive probing)

2. Encode relational structure: $e_i = \text{encode}(o_i)$
3. Compute curvature: $\Delta(i,j,k) = d(i,j) + d(j,k) + d(k,i)$
4. Canonicalize: $\Delta_c = C(\Delta)$
5. Compute kappa signature: $\kappa = \kappa(\Delta_c)$
6. Compute structural hash: $h = H(\Delta_c, \kappa)$

No stage feeds back into a previous stage. The pipeline is a DAG.

VI. kappa Signature (Structural Stability)

kappa maps the canonical curvature to a normalized stability distribution:

$$\kappa : \Delta_c \rightarrow \mathbb{R}^n$$

$$\kappa_i = ||e_i - \mu|| / \sum_j ||e_j - \mu||$$

Properties:

- Deterministic: identical Δ_c produces identical κ
- Order-independent: no dependence on index labeling
- Canonicalization-invariant: $\kappa(\Delta_c) = \kappa(C(\Delta))$
- No optimization: kappa is a structural observable, not a target

kappa is not a scalar distance from a boundary. It is a distribution that characterizes the structural position of the field within the admissible domain D .

VII. Topological Hashing

Structural fingerprint:

$$H(\Phi) = H(\Delta_c, \kappa)$$

Topology-extended fingerprint:

$$H_{\text{topo}}(\Phi) = H(\Delta_c \mid dD, \text{invariants})$$

Requirements:

- $H(\Phi_1) = H(\Phi_2)$ iff $\Phi_1 \text{ equiv } \Phi_2$
- Sensitivity to Delta deformations
- Invariance under representation (orbit-equivalent fields produce identical H)

Implementation:

$$h_{256} = \text{SHA256}(\text{serialize}(\Delta_c, \kappa, \text{LTL}))$$

$$h_3 = \text{SHA3-256}(\text{serialize}(\Delta_c, \kappa, \text{LTL}))$$

VIII. Epistemic Merkle Structure

For a field trajectory T :

$$T = (\Phi_0 \rightarrow \Phi_1 \rightarrow \dots \rightarrow \Phi_n)$$

Define per-state hashes:

$$h_i = H(\Phi_i)$$

Merkle root:

$$M(T) = \text{Merkle}(h_0, h_1, \dots, h_n)$$

Trajectory validity conditions:

- For all i : Δ_i in D (representability)
- For all i : $K(\Phi_i) \geq \kappa$ (coherence preserved)
- Hashes consistent with Merkle tree structure

The Merkle structure enables verification of individual trajectory segments without requiring full trajectory reconstruction. Any party with $M(T)$ can verify any h_i in $O(\log n)$.

IX. Non-Intervention Principle

EK enforces the structural invariant:

$$d\Phi / dEK = 0$$

The system:

- Does not modify the observed system
- Introduces no feedback into the observed system
- Performs no optimization on the observed system
- Encodes no semantic content in its outputs

Removal invariance:

$$\text{If EK is removed: } \Phi \text{ remains unchanged.}$$

EK is observational and non-causal.

X. Implementation Correspondence

Module structure:

- `representability.py` -> verification of Δ in D
- `canonical.py` -> orbit canonicalization via C

```
kappa.py -> kappa signature computation
hash.py -> SHA256 + SHA3-256 structural hashing
merkle.py -> trajectory verification
pipeline.py -> full EK 2.0 flow
```

CI enforcement:

Phi Integrity Test Suite verifies invariants on every commit:
Delta in $D \mid C$ invariant $\mid d\Phi/dEK = 0 \mid$ determinism

XI. Properties

EK 2.0 satisfies:

Deterministic: identical inputs produce identical outputs
Non-adaptive: probe set is fixed before execution
Non-interventional: $d\Phi/dEK = 0$
Representation-invariant: $C(g \cdot \Delta) = C(\Delta)$ for all g in S_8
Topology-sensitive: H distinguishes non-equivalent field configurations

XII. Conclusion

Epistemic Cryptography 2.0 defines a framework where:

structure is observable,
identity is canonical,
verification is deterministic,
and no intervention is required.

Appendix A — Core Constraints

Δ in $\text{Im}(d)$ — curvature lies in image of exterior derivative
 R in $\mathfrak{so}(8)$ — antisymmetric relational matrix
 C invariant under S_8 — canonical form is orbit-invariant
 $d\Phi / dEK = 0$ — non-intervention invariant

Appendix B — Design Principle

The system does not decide. It selects.

The system does not optimize. It canonicalizes.

The system does not interpret. It preserves structure.

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